

Drone with volca keys

Manual for the Deluge with firmware 1.2.1-mastertune v16 (or newer) and a Korg volca keys

The volca keys is a small analog synthesizer with three oscillators, one filter and one envelope. Played from the Deluge, it delivers long, warm tones that move slowly. The Deluge's **Frequency Drone** lays exact tones and beatings underneath: binaural, monaural or isochronic. This manual shows you how to wire, tune, play and record both.

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|-------------------------------------|--------------------------------------|
| 1. What the combination is good for | 7. Together with the Frequency Drone |
| 2. Wiring | 8. The volca through the Deluge |
| 3. Setting up the volca keys | 9. Recipes |
| 4. Setting up the Deluge | 10. CC table |
| 5. Tuning: volca and Master Tune | 11. Troubleshooting |
| 6. Playing drones on the volca | 12. Limits and open points |

Start quiet. Sustained tones, especially low ones, can seem louder than expected on headphones. Start with a low volume on the volca and the Deluge and raise it slowly.

Sources: The information on the volca keys is based on Korg's MIDI chart (v1.02), excerpts from the Korg manual, reviews and forums. Points marked **verify** are uncertain or have conflicting sources. Check them on the device or in the Korg manual before you rely on them.

1. What the combination is good for

What the volca keys adds

- Analog sound that drifts slightly and so feels alive
- Three oscillators: as a chord of up to three notes (Poly) or as one wide tone (Unison, Octave, Fifth)
- Filter, LFO and delay for slow movement
- Remote control via MIDI: notes, pitch bend and 16 control changes (CC)

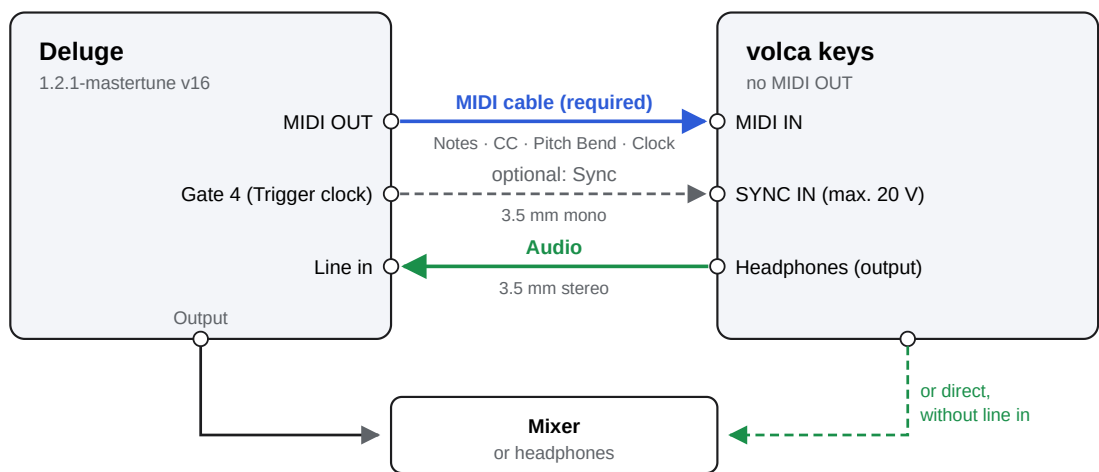
What the Deluge adds

- Long notes in a MIDI clip, CC automation, MIDI clock
- The Frequency Drone with exact tones: Tone, Binaural, Monaural, Isochronic, with sidechain
- Drone tracks (v16) that you launch like clips
- Recording the volca via the line in, also as an audio clip that follows the Master Tune

Typical uses: an analog pad over a binaural foundation, a filter sweep over many bars, a slow beating between the drifting analog tone and the exact drone tone, or a recording of the volca that you then retune to 432 Hz.

Rule of thumb: The volca keys is fixed at A = 440 Hz. Leave the Master Tune at **440.0** and set the drone tones to **Note**. Then the volca and the drone match. For 432 Hz and other tunings, see chapter 5.

2. Wiring



Basic setup. Only the MIDI cable is required. Sync and line in are optional. Without line in, the volca goes straight to the mixer.

Connection	Cable	Purpose
Deluge MIDI OUT → volca MIDI IN	MIDI cable. That the volca has a 5-pin DIN socket is very likely verify	Notes, pitch bend, CCs and MIDI clock. This is the only connection you need.
Deluge Gate 4 → volca SYNC IN	3.5 mm mono	Only if the volca's sequencer should run from pulses. For this, Gate 4 must be set to <i>Clock</i> (chapter 4). Not needed for notes from the Deluge. SYNC IN takes at most 20 V.
volca headphones → Deluge line in	3.5 mm stereo, plug to fit the line-in socket	Recording as an audio clip, input as an oscillator, resampling (chapter 8)
or volca headphones → mixer	3.5 mm stereo	if the volca should not run through the Deluge

- **No return path:** The volca keys has no MIDI OUT (only as a mod). So the Deluge cannot record knob movements on the volca as MIDI. Automation is always created on the Deluge.
- The volca's **SYNC OUT** (3.5 mm mono, 5 V, 15 ms pulse) is not needed in this setup.
- **Power:** 6 × AA last about 10 hours. For long drone sessions a 9 V power adapter (DC) is safer.

3. Setting up the volca keys

According to the Korg manual, you reach the volca's global settings by **switching it on while holding the FUNC button**. The parameters are then on the step buttons: **1** Auto Power Off, **2** battery type, **3** SYNC OUT polarity, **4** SYNC IN polarity, **6** MIDI clock, **7** MIDI ShortMessage. **REC** saves and restarts, **PLAY** cancels without saving.

Setting	For this setup	Remark
MIDI channel	leave at 1 , the default	The Deluge clip then also goes to channel 1. The sources differ on exactly how to change the channel: buttons 1–16 directly, via MEMORY or with FUNC + button 16 verify
MIDI clock	LED of step 6 on = Auto (follows MIDI clock), off = internal	Only matters if the volca's sequencer or Motion Sequence should run at the Deluge's tempo. Under MIDI clock, the volca's tempo knob has no effect.

Setting	For this setup	Remark
SYNC IN polarity	rising edge (default)	global parameter on button 4; only matters with a sync cable
Auto Power Off	off for long drones	Button 1. Otherwise the volca switches itself off after about 4 hours.
MIDI ShortMessage	leave at default	Button 7. What exactly the parameter does is not documented verify . If the volca does not respond to notes or CCs from the Deluge, switch it.

If a cable is plugged into SYNC IN, the volca's sequencer follows only the pulses and ignores its own clock. The Korg manual also lists the tempo range as a global parameter. It plays no role in this setup, because the tempo comes from the Deluge.

4. Setting up the Deluge

MIDI clip for the volca

Setting up a MIDI clip is part of the standard operation of 1.2.1. The mastertune documents do not describe it; the official Deluge manual explains it in detail. In short:

1. Create a new clip as usual and switch to clip view.
2. Press the [MIDI](#) button. The clip now plays to a MIDI channel, and the MIDI LED lights up.
3. Turn Select until the volca's channel appears (default: 1).
4. Enter notes. For drones: long notes, see chapter 6.

Clock and tuning

Menu	What you set
Settings → MIDI → Clock → Output Title "MIDI clock out"	On if the volca should run along at the tempo (its MIDI clock set to Auto). Since v15 the MIDI clock output also keeps running when the Deluge itself follows an external MIDI clock.
Settings → Gate → Gate output 4 → Clock Title "Gate out4 mode" Settings → Trigger clock → Output → PPQN Title "Output PPQN"	Only with a sync cable. The trigger clock comes from Gate 4 , but only if its mode is set to Clock . The default is V-trig, and then the sync cable stays silent. Set <i>Output PPQN</i> to 2 : the sources point to 1 pulse per 2 steps verify . Compare the volca's tempo with the Deluge's. MIDI clock without a sync cable is simpler.
Settings → Tuning → Master tune (Hz) 7-segment: TUNE → MTUN	Leave it at 440.0 for the volca (chapter 5). Range 415.3 to 466.2 Hz in 0.1 Hz steps.

5. Tuning: volca and Master Tune

The volca keys has **neither a Master Tune nor a fine tuning**. So A = 440 Hz is effectively fixed. The Deluge's Master Tune therefore shifts everything in the Deluge, but not the volca.

Sound source	Follows the Master Tune?
Drone tones as Note	yes
Drone tones in Hz	no, the frequency stays fixed
Deluge synths, kits, samples	yes

Sound source	Follows the Master Tune?
Audio clips, including recordings of the volca	yes, by time-stretch; the tempo stays in sync. A recording is only shifted by the difference from the tuning it was recorded at, never twice.
volca, played via MIDI	With a tuning other than 440 Hz, the Deluge sends RPN 1 (Channel Fine Tuning) on its channel. That the volca evaluates RPN 1 is not documented and, given the missing fine tuning, unlikely verify . Expect it to stay at 440.
volca via line in as an oscillator	no, it sounds unchanged in its own tuning

The volca's auto-tuning

The volca retunes itself: after about **10 seconds of silence** it corrects the drift of its oscillators. This runs continuously; no warm-up time is needed. For drones this means [verify](#): as long as a sustained tone sounds, it does not retune and can drift slightly over time. Give it more than 10 seconds of silence between two pieces, for example by stopping with [PLAY](#) on the Deluge.

At 440 Hz (recommended)

Master Tune at 440.0, and all drone tones that should match the volca set to **Note** (press the upper gold knob or use *Pitch as* in the menu). This is the approach from chapter 13 of the Drone manual: tune the Master Tune to the external device. For the volca that means 440.0. If the Master Tune has never been set to anything else since power-on, the Deluge also sends no RPN 1.

At 432 Hz: honest options

From 440 to 432 Hz is −31.77 cents. The volca cannot do this by itself. There are three ways:

Way	How to	Drawback
A. Record, then retune (safest way)	Master Tune at 440.0 , record the volca as an audio clip (chapter 8). Then Master Tune to 432.0: the clip sounds 31.77 cents lower, the tempo stays. Drone tones set as notes follow.	Not live. Time-stretch needs more CPU time and has very quiet artifacts. Important: A recording made at Master Tune 432.0 counts as 432 material and would stay at the volca's pitch, that is 440.
B. Pitch bend	Lower the volca by about 32 cents with pitch bend. With a bend range of ± 2 semitones, that is about 16 % of the way down (14-bit value about 6890 instead of 8192). Check with a tuner.	The bend range comes from a forum. Resolution and interaction with the auto-tuning are not documented verify . How the Deluge holds a fixed bend value in a MIDI clip is not in the mastertune documents (official manual). Any further bend adds on top.
C. Deliberately against each other	Master Tune 432.0, the volca stays at 440. The same notes then beat audibly against each other: at A2 by 4 Hz, at A3 by 8 Hz.	Only useful as an effect, not as a tuning.

A drone tone at **432 Hz in Hz** is not the same as Master Tune 432.0: it stays fixed, even if you change the Master Tune. At Master Tune 432.0 the note A3 gives exactly 432 Hz.

6. Playing drones on the volca

Long notes

- Enter long notes in the MIDI clip, for a pad across the whole clip. The volca holds the tone as long as the note lasts.
- **A new note restarts the envelope for all held notes** (secondary source). So for a calm pad, start all notes of a chord at the same time. Staggered entries create a new attack each time.
- A note that starts at step 1 and fills the whole clip is held by the Deluge across the loop boundary. It sends neither Note Off nor Note On there, so the volca does not retrigger.
- In Poly mode the volca plays up to three notes. What happens with more than three notes is not documented [verify](#). Stick to three at most.

Voice modes (VOICE)

Mode	Sound	For drones
Poly	3 notes, sawtooth	chord of up to three notes, for example root, fifth, octave
Unison	mono, all three oscillators on one note	wide, carrying single tone; beating with Detune
Octave	mono	single tone with an octave sound
Fifth	mono	single tone with a fifth sound, open and empty
Unison Ring	square wave with ring modulation	metallic, bell-like tone
Poly Ring	square wave with ring modulation	metallic chord

Via MIDI, CC 40 selects the mode. Which values give which mode is not documented [verify](#). So set the mode with the volca's knob. The same applies to the octave setting (CC 41, 6 octaves on the knob).

Envelope for sustained tones

- **Attack** high: the tone fades in slowly.
- **Sustain** fully up: the tone holds its volume. That it then holds indefinitely is inferred [verify](#).
- **Decay/Release** long: the tone fades out softly.
- There is only **one** envelope and **one** filter (12 dB) for all three voices.

Detune and portamento

- **Detune** (CC 42) detunes the oscillators against each other. In Unison this creates an inner beating that makes the tone wide.
- **Portamento** (CC 5) makes the tone glide from note to note. With long notes this becomes a slow drift.

Slow filter sweeps

- **Via CC:** automate CC 44 (Cutoff) in the MIDI clip over many bars. How to record or draw CCs in a MIDI clip is part of the standard operation of 1.2.1 (official manual). The volca has no MIDI OUT, so the route always goes through the Deluge.
- **Via LFO:** LFO Rate (CC 46) low, LFO Cutoff Int (CC 48) to taste. Waveforms sawtooth, triangle, square. The LFO can be synced to the start of the note (Trigger Sync).
- **Steps:** according to a review, the filter steps audibly at high resonance. For very slow sweeps, keep the resonance moderate.
- **Delay:** Delay Time (CC 52) and Feedback (CC 53) add space. Feedback also sets the amount of delay in the sound.

7. Together with the Frequency Drone

You open the song drone in song view with **SCALE** or via **SELECT** → song menu → *Drone*. The Drone manual describes how to operate it. This chapter is about how the two play together.

Hitting the same notes

- Set the drone tone to **Note**: press the upper gold knob or use *Pitch* as in the menu. Turning the upper gold knob selects the note, with **SHIFT** the cents.
- The note names follow the Deluge numbering: A3 = 440 Hz, middle C = C3. Because the Deluge sends the volca its notes, this numbering applies equally to the MIDI clip and the drone.
- For a foundation, place the drone one or two octaves below the volca's root note. For warmth, a fifth or octave above works well.

Binaural beat under an analog chord

The volca plays a chord in Poly mode. Below it are one or two drone tones in **Binaural** mode (**KIT** button) on the root note, one or two octaves lower. The beating arises only in the drone; the volca provides the pad above it. This works **only with headphones**.

Beating between analog and digital

If you put a drone tone on the same note as the volca and detune it by a few cents, the two tones slowly beat against each other. The beating is about $frequency \times cents \div 1731$: at A2 (220 Hz) and 3 cents, about 0.4 Hz. Because the volca drifts slightly, the beating drifts with it. This sounds alive.

Sidechain

The drone can duck under the song's sidechain triggers, for example under a kick with *Send to sidechain*. In the drone menu under *Sidechain* you find *Volume ducking*, SYNC, ATTACK/RELEASE and SHAPE. The volca does not duck along; its pad stays steady while the drone pumps underneath.

Drone tracks (v16)

In the drone view, **SHIFT** + **KIT** creates a kit DRONE1 (DRONE2 ...) from the drone tones. The rows' notes are the gate; a row's arpeggiator makes the tone pulse. You launch the clip in song view together with the volca's MIDI clip. That way the volca and the drone start and end together.

Level: a drone tone at Level 50 and Volume 50 sits at -12 dBFS. The volca comes in via the line in or the mixer. Balance the two by ear and start with a drone Level of about 30.

8. The volca through the Deluge

With a cable from the volca's headphone output into the Deluge's line in, you can record the volca or send it through the Deluge. Start with a low volume on the volca. The official Deluge manual describes in detail how to use audio clips and recording. The menu items are called *Audio source*, *Input (left/right/stereo)* and *Monitoring*.

Route	What happens	Tuning
Audio clip	An audio track with <i>Audio source</i> set to the input records the volca. With <i>Monitoring</i> you hear the input while recording.	The recording then follows the Master Tune. If the tuning is not 440 while recording, the WAV file gets an mtun block and is never tuned twice.

Route	What happens	Tuning
Input as oscillator	A Deluge synth uses the line in as an oscillator. This way the volca runs through the Deluge's filters, envelopes and effects.	does not follow the Master Tune, because the volca itself is already tuned
Resampling	The Deluge records its own output, with the volca via the line in.	like an audio clip, with an mtun block if Master Tune $\neq$ 440

- **Drone in recordings:** the song drone can be heard via the output and USB audio and also ends up in **resampling recordings**. It is **missing from stems**.
- **432 Hz:** for way A from chapter 5, record at Master Tune 440.0 and only change it afterwards.

9. Recipes

Values on the volca are knob positions by ear. The Deluge values come from the Drone manual. All recipes assume MIDI channel 1 and Master Tune 440.0, except recipe 5.

1. Analog chord over a binaural base

1. volca: VOICE to **Poly**, Attack about half, Sustain fully up, Decay/Release long, Cutoff half.
2. Deluge: MIDI clip of 4 bars, with C3, E3 and G3 together across the whole clip.
3. **SCALE** in song view. Tone 1: mute pad on, **KIT** (Binaural), *Pitch as Note*, note C2, Timbre Sine, Beat 6 Hz, Level 32.
4. Tone 2: Binaural, note G2, Timbre Soft, Beat 6 Hz, Level 26.
5. **PLAY**. Listen on headphones.

2. Unison drone with a filter sweep over 16 bars

1. volca: VOICE to **Unison**, Detune about a quarter, Attack long, Sustain fully up, resonance moderate.
2. Deluge: MIDI clip of 16 bars, one note A1 across the whole clip.
3. Automate CC 44 (Cutoff) in the clip: from low to high over 8 bars, back over 8 bars.
4. Drone tone 1: mode Tone (**SYNTH**), *Pitch as Note* (press the upper gold knob), note A0, Timbre Sine, Level 30. It carries the low end while the filter moves.

3. Slow beating against the volca

1. volca: **Unison**, Detune fully down, Sustain fully up. MIDI clip with one long note A2.
2. Drone tone 1: mode Tone, Timbre Soft, *Pitch as Note* (press the upper gold knob), note A2, then turn **SHIFT** + upper gold knob to +3 cents.
3. Set Level so that the drone and the volca are about equally loud. This creates a beating of about 0.4 Hz.
4. Optional: Life 10 (Delay section, upper gold knob) for extra movement.

4. Pulsing ring drone in tempo

1. volca: VOICE to **Poly Ring**, Sustain fully up, Delay Time medium, Feedback low.
2. Deluge: MIDI clip with two long notes a fifth apart, for example D2 and A2.
3. Drone tone 1: **CV** (Isochronic), *Pitch as Note* (press the upper gold knob), note D1, Timbre Soft. Press the lower gold knob (SYNC) and turn it to 1/8T. Attack/Release section: Attack 20, Release 30.
4. Add a kick with *Send to sidechain*, and set *Volume ducking* to 20 in the drone menu.

5. Bringing the volca to 432 Hz

1. Settings → Tuning → Master tune (Hz) to **440.0**.
2. Set up the volca as in recipe 2 and record it via the line in into an audio clip, for example 8 bars.
3. Master Tune to **432.0**. The audio clip now sounds 31.77 cents lower; the tempo stays.
4. Drone tones set as notes follow; A3 is now at 432 Hz. Tones in Hz stay fixed.
5. Do not play the volca live along with it any more: it is still at 440.

10. CC table

Based on Korg's MIDI chart (v1.02). By the MIDI standard, CC values range from 0 to 127. According to Korg, some parameters can only be reached via MIDI (marked with * in the chart). The Korg chart itself shows which ones.

CC	Parameter	For drones
5	Portamento	gliding between long notes
11	Expression	check the effect on the device verify
40	Voice	voice mode; value ranges not documented, choose it on the knob verify
41	Octave	octave setting; value ranges not documented verify
42	Detune	width and inner beating, especially in Unison
43	VCO EG Int	pitch follows the envelope; use little for calm drones
44	Cutoff	the knob for slow filter sweeps
45	VCF EG Int	how far the envelope opens the filter
46	LFO Rate	low for slow movement
47	LFO Pitch Int	vibrato; a little makes the tone come alive
48	LFO Cutoff Int	the filter breathes with the LFO
49	Attack	fade in
50	Decay/Release	fade out
51	Sustain	fully up for sustained tones
52	Delay Time	space
53	Delay Feedback	repeats and amount of delay
Other MIDI messages		Status
Note On/Off		yes
Velocity		evaluated, according to a forum verify
Pitch Bend		yes; range ± 2 semitones according to a forum, not changeable verify
Program Change, RPN 1 (Fine Tuning), Local Control		not documented, probably not available verify

The list contains no CC for resonance. The Frequency Drone itself sends no MIDI, takes no notes and cannot be automated. All CCs to the volca come from the MIDI clip.

11. Troubleshooting

Problem	Cause and solution
The volca does not play	Cable in the Deluge's MIDI OUT and the volca's MIDI IN? Clip on the same channel as the volca (default 1)? Clip launched and not muted? The volca's volume turned up? Switch the global parameter MIDI ShortMessage (button 7) verify .
The volca's tempo knob does nothing	It follows the MIDI clock (LED of step 6 on) or a sync cable. This is intended.
The volca's sequencer runs twice or half as fast	With a sync cable: <i>Output PPQN</i> does not match. Try 4 instead of 2, or use MIDI clock without a cable. If it does not run at all: Gate 4 set to <i>Clock</i> ?
The pad sound restarts	A new note restarts the envelope for all notes. Start chord notes at the same time. At the loop boundary: the note must start at step 1 and fill the whole clip.
A note is missing from the chord	More than three notes at once, or a mono mode is active. VOICE to Poly, at most three notes.
The volca and the drone do not match	Master Tune not at 440.0? The volca does not follow it. Drone tones in Hz instead of as notes? After playing for a long time the volca can drift: give it more than 10 s of silence, then it retunes.
A recording sounds too high at 432 Hz	It was made at Master Tune 432.0 and so counts as 432 material. Record it again at 440.0.
A filter sweep sounds steppy	At high resonance the filter steps audibly. Lower the resonance or make the movement with the LFO.
The volca suddenly switches off	Auto Power Off after about 4 hours, or empty batteries. Turn off the parameter on button 1, use a power adapter.

12. Limits and open points

Limits

- The volca keys has no fine tuning. The Deluge's Master Tune does not retune it; 432 Hz only works via recording or pitch bend (chapter 5).
- Paraphonic: three voices share one filter and one envelope. There are no independent envelopes per note.
- No MIDI OUT: the volca can send neither notes nor knob movements to the Deluge.
- Auto-tuning only in pauses: with sustained tones lasting hours, the volca can drift slightly.

Open points [verify](#)

Not documented, or with conflicting sources: the MIDI IN socket as 5-pin DIN (very likely); the exact way to set the MIDI channel; the pulses per step at SYNC IN (probably 2 PPQN); the value ranges of CC 40 and CC 41, the effect of CC 11 and which parameters can only be reached via MIDI; pitch bend range, velocity, Program Change, RPN 1 and Local Control; the behavior with more than three notes and indefinite holding with Sustain fully up; whether a sustained tone suppresses the auto-tuning. On the Deluge side: a fixed pitch bend value in a MIDI clip.

Sources on the volca: Korg's MIDI chart v1.02, product page korg.com (connections, power), excerpts from the Korg manual, Sound On Sound, Sweetwater, sonicstate, zZounds, korgforums.com. On the Deluge: README and Drone manual of 1.2.1-mastertune, source code v16 (menu names, Gate 4 as trigger clock and its mode, long notes at the loop boundary, drone during resampling).